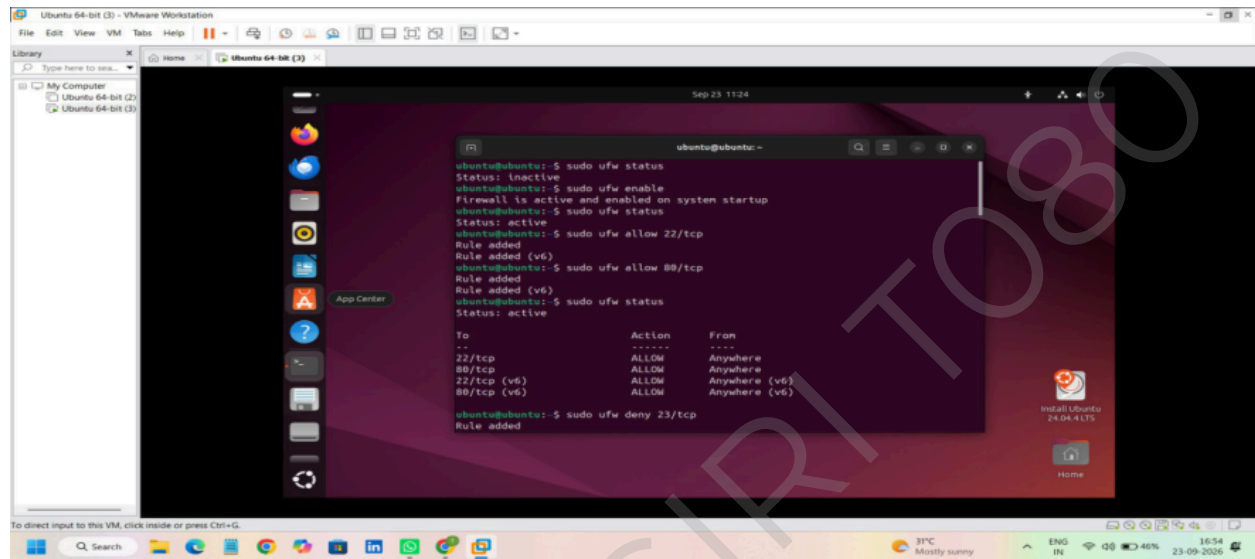


SHETH LUJ AND SIR MV COLLEGE

Aim:- Configure and test firewall rules to control network traffic, filter packets based on specified criteria, and protect network resources from unauthorized access.

Step 1: Open Ubuntu Terminal



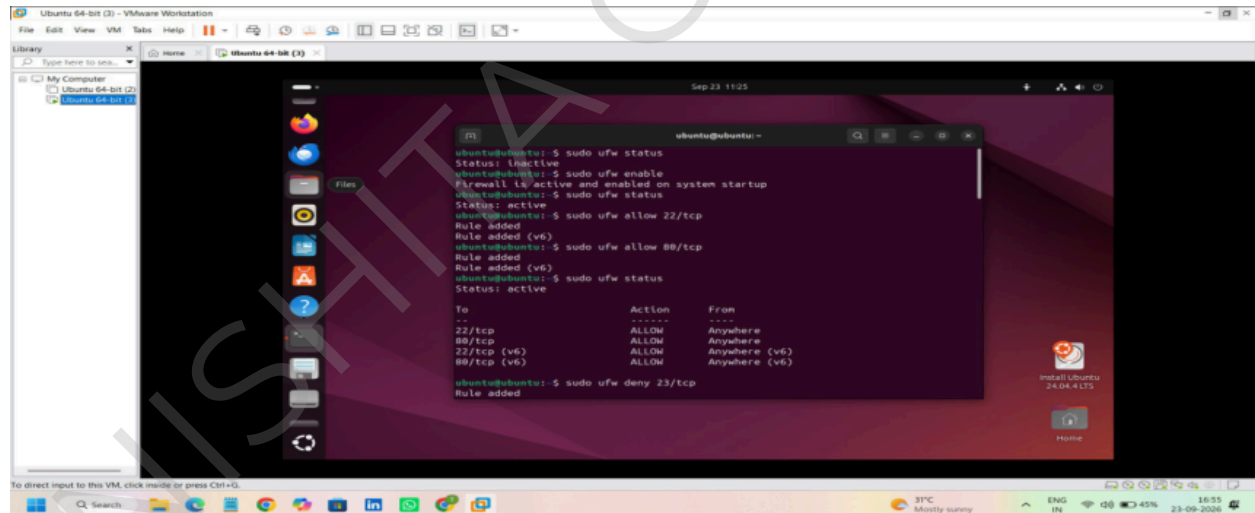
The screenshot shows a terminal window within a VMware Workstation environment. The terminal displays the following commands and output:

```
ubuntu@ubuntu:~$ sudo ufw status
Status: inactive
ubuntu@ubuntu:~$ sudo ufw enable
Firewall is active and enabled on system startup
ubuntu@ubuntu:~$ sudo ufw status
Status: active
ubuntu@ubuntu:~$ sudo ufw allow 22/tcp
Rule added
ubuntu@ubuntu:~$ sudo ufw allow 80/tcp
Rule added
ubuntu@ubuntu:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
80/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
80/tcp (v6) ALLOW Anywhere (v6)

ubuntu@ubuntu:~$ sudo ufw deny 23/tcp
Rule added
```

Step 2: Enable the Firewall



The screenshot shows a terminal window within a VMware Workstation environment. The terminal displays the following commands and output:

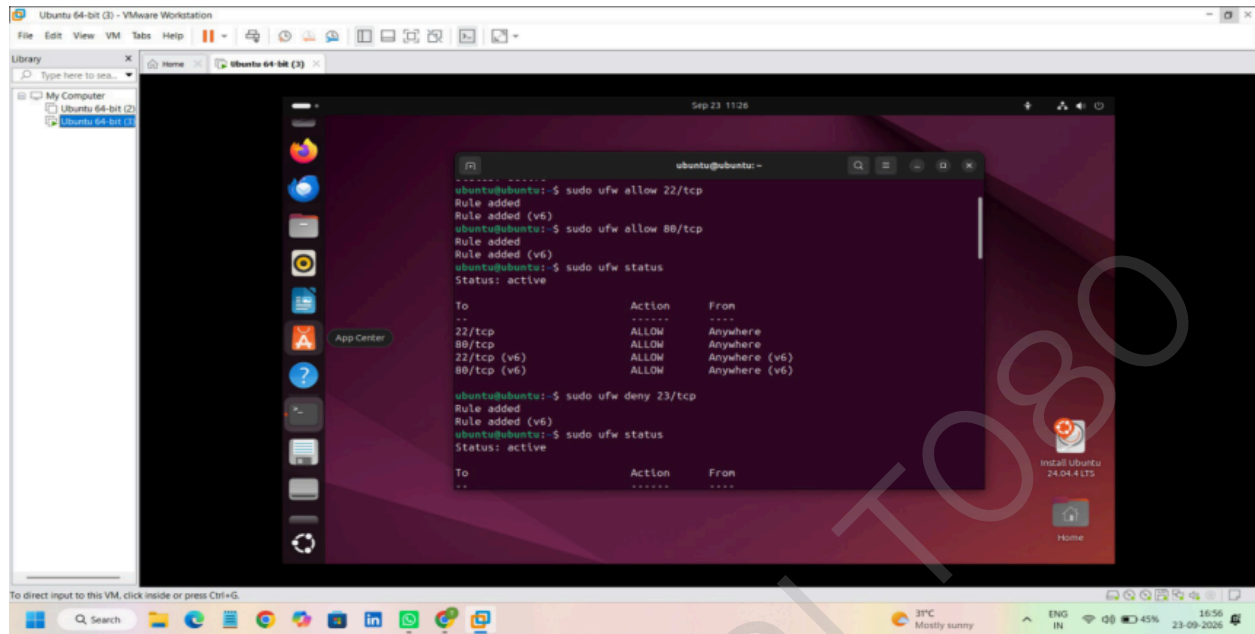
```
ubuntu@ubuntu:~$ sudo ufw status
Status: inactive
ubuntu@ubuntu:~$ sudo ufw enable
Firewall is active and enabled on system startup
ubuntu@ubuntu:~$ sudo ufw status
Status: active
ubuntu@ubuntu:~$ sudo ufw allow 22/tcp
Rule added
ubuntu@ubuntu:~$ sudo ufw allow 80/tcp
Rule added
ubuntu@ubuntu:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
80/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
80/tcp (v6) ALLOW Anywhere (v6)

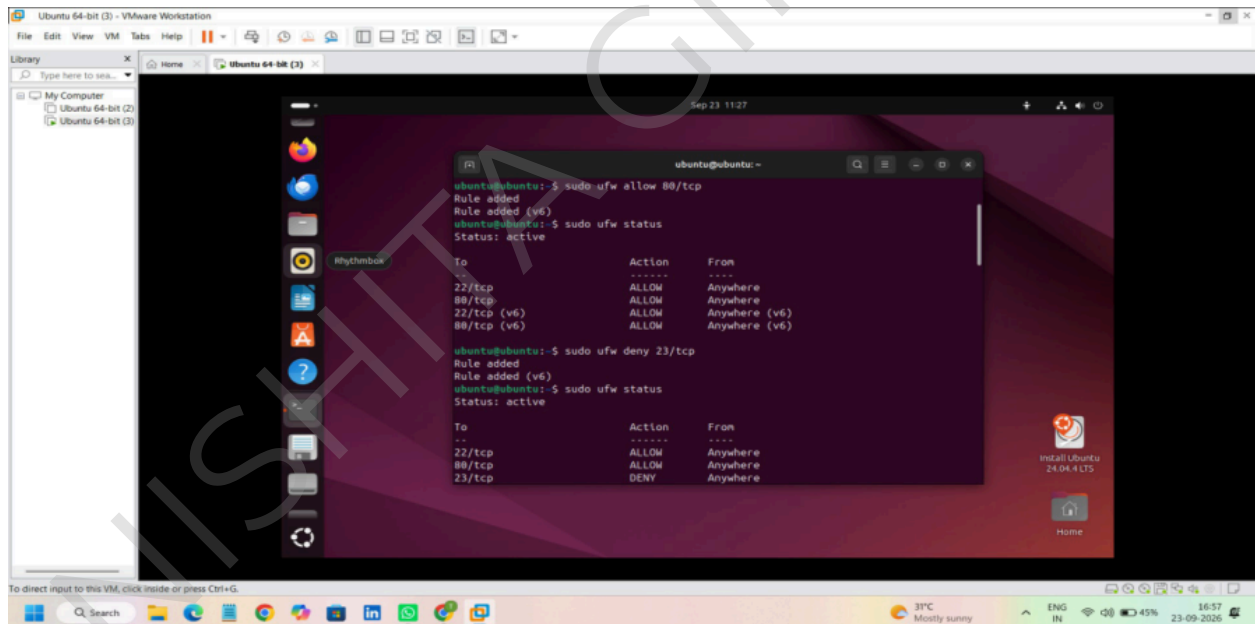
ubuntu@ubuntu:~$ sudo ufw deny 23/tcp
Rule added
```

SHETH LUJ AND SIR MV COLLEGE

Step 3: Allow SSH Traffic

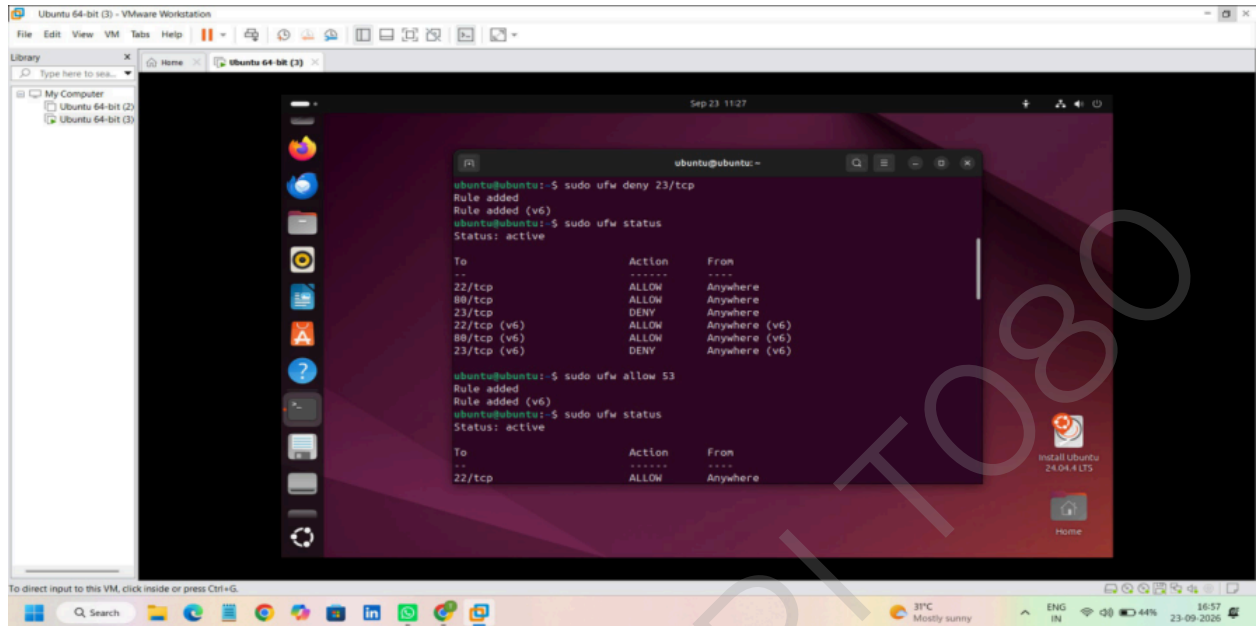


Step 4:- Step 4: Allow HTTP Traffic



SHETH LUJ AND SIR MV COLLEGE

Step 5: Block a Port



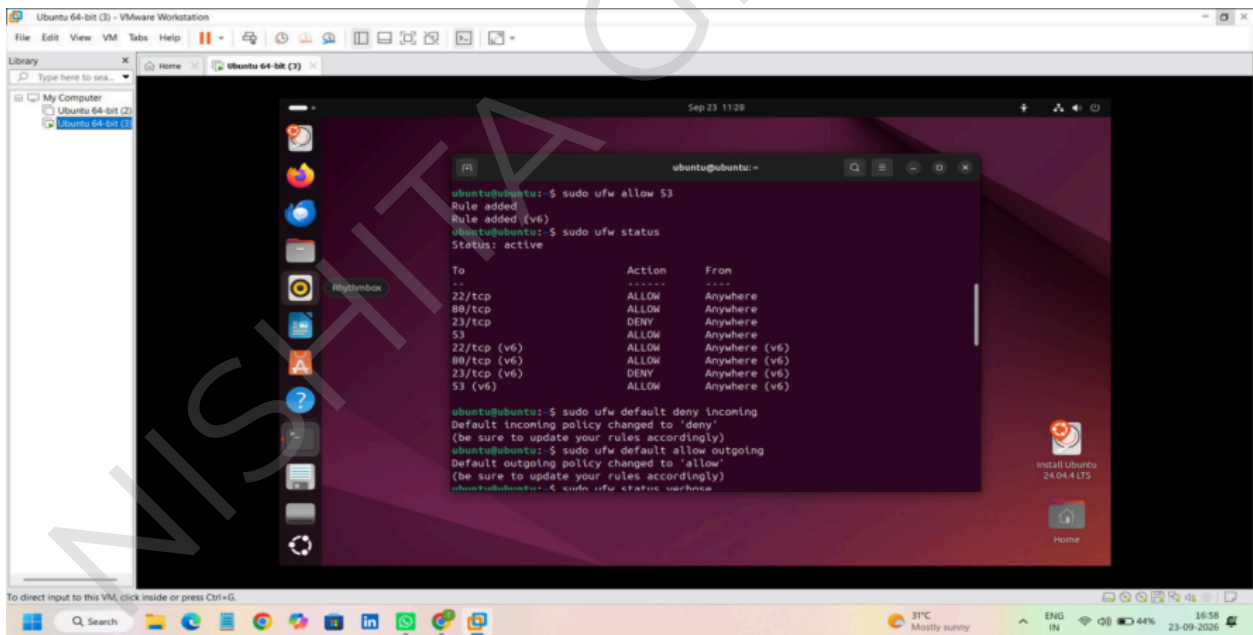
```
ubuntu@ubuntu:~$ sudo ufw deny 23/tcp
Rule added
Rule added (v6)
ubuntu@ubuntu:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
80/tcp ALLOW Anywhere
23/tcp DENY Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
80/tcp (v6) ALLOW Anywhere (v6)
23/tcp (v6) DENY Anywhere (v6)

ubuntu@ubuntu:~$ sudo ufw allow 53
Rule added
Rule added (v6)
ubuntu@ubuntu:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
```

Step 6: Allow DNS Traffic



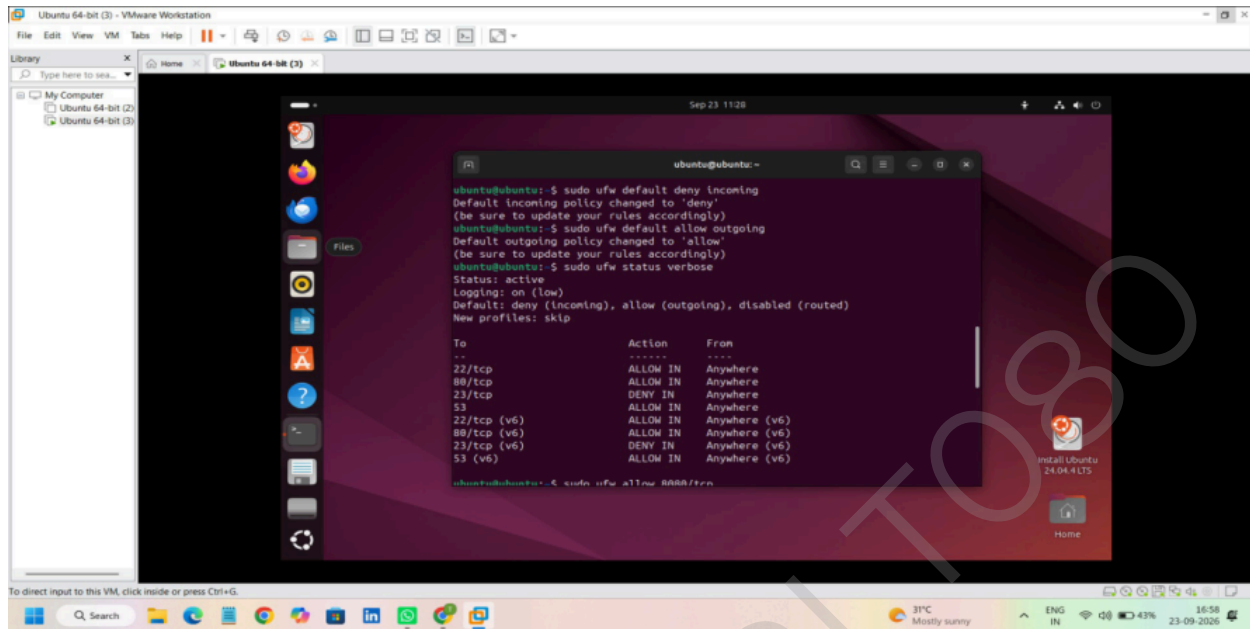
```
ubuntu@ubuntu:~$ sudo ufw allow 53
Rule added
Rule added (v6)
ubuntu@ubuntu:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
80/tcp ALLOW Anywhere
23/tcp DENY Anywhere
53 ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
80/tcp (v6) ALLOW Anywhere (v6)
23/tcp (v6) DENY Anywhere (v6)
53 (v6) ALLOW Anywhere (v6)

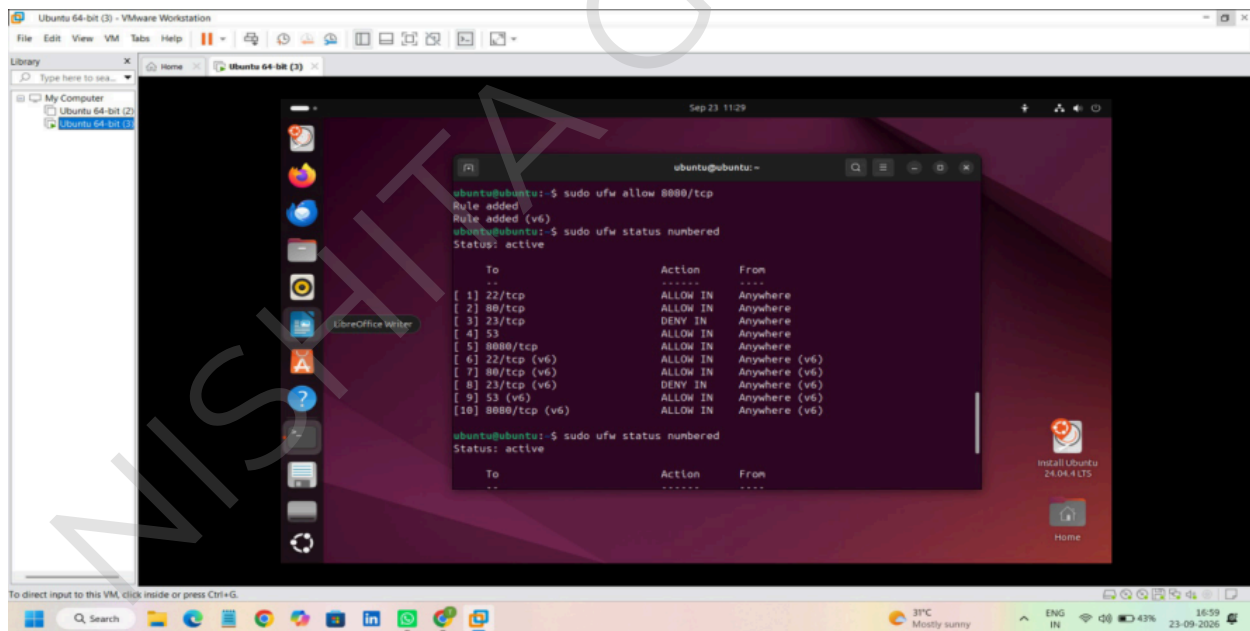
ubuntu@ubuntu:~$ sudo ufw default deny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
ubuntu@ubuntu:~$ sudo ufw default allow outgoing
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)
ubuntu@ubuntu:~$ sudo ufw status verbose
```

SHETH LUJ AND SIR MV COLLEGE

Step 7: Set Default Firewall Policies

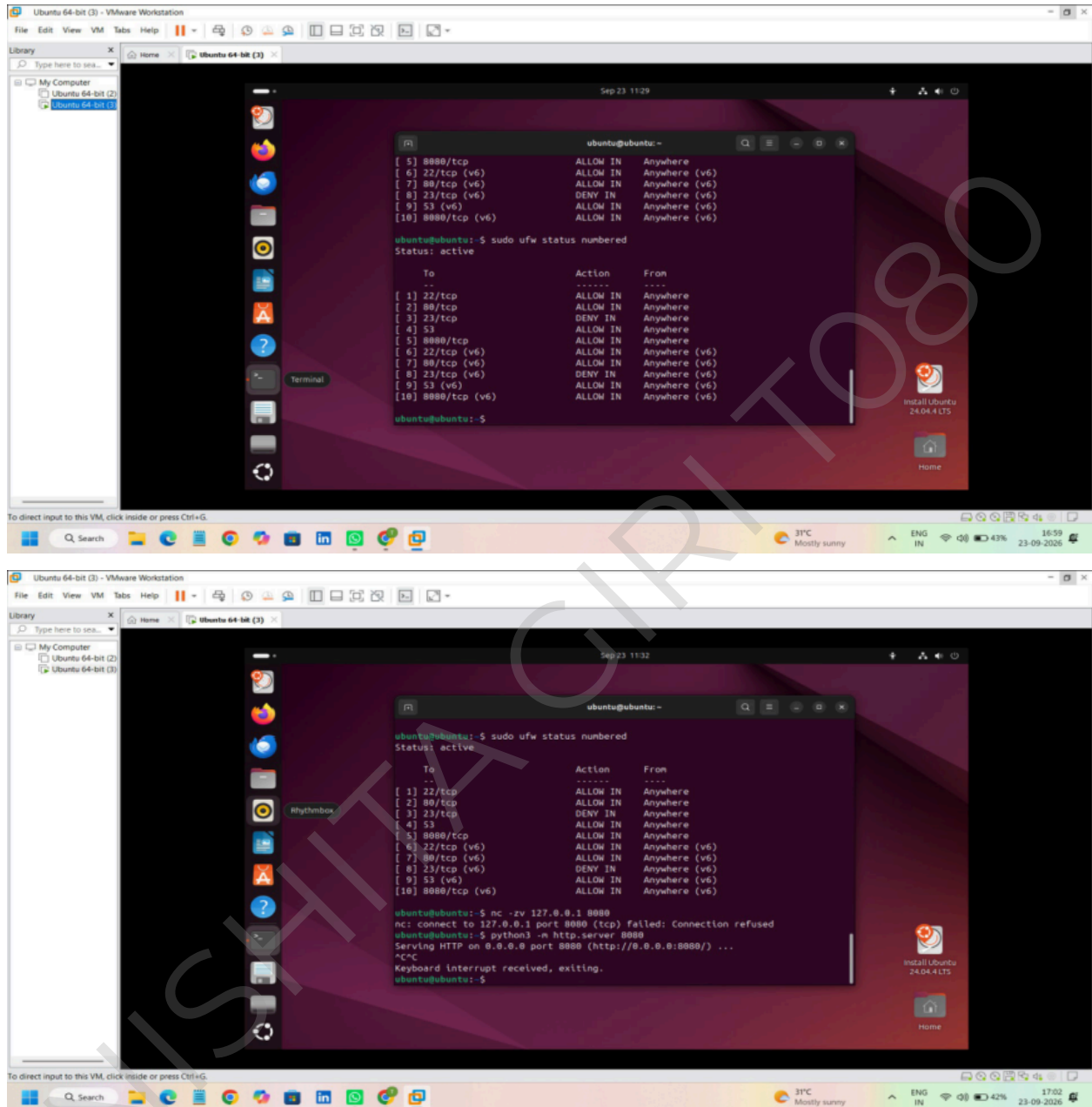


Step 8: Create a Rule for a Specific Port



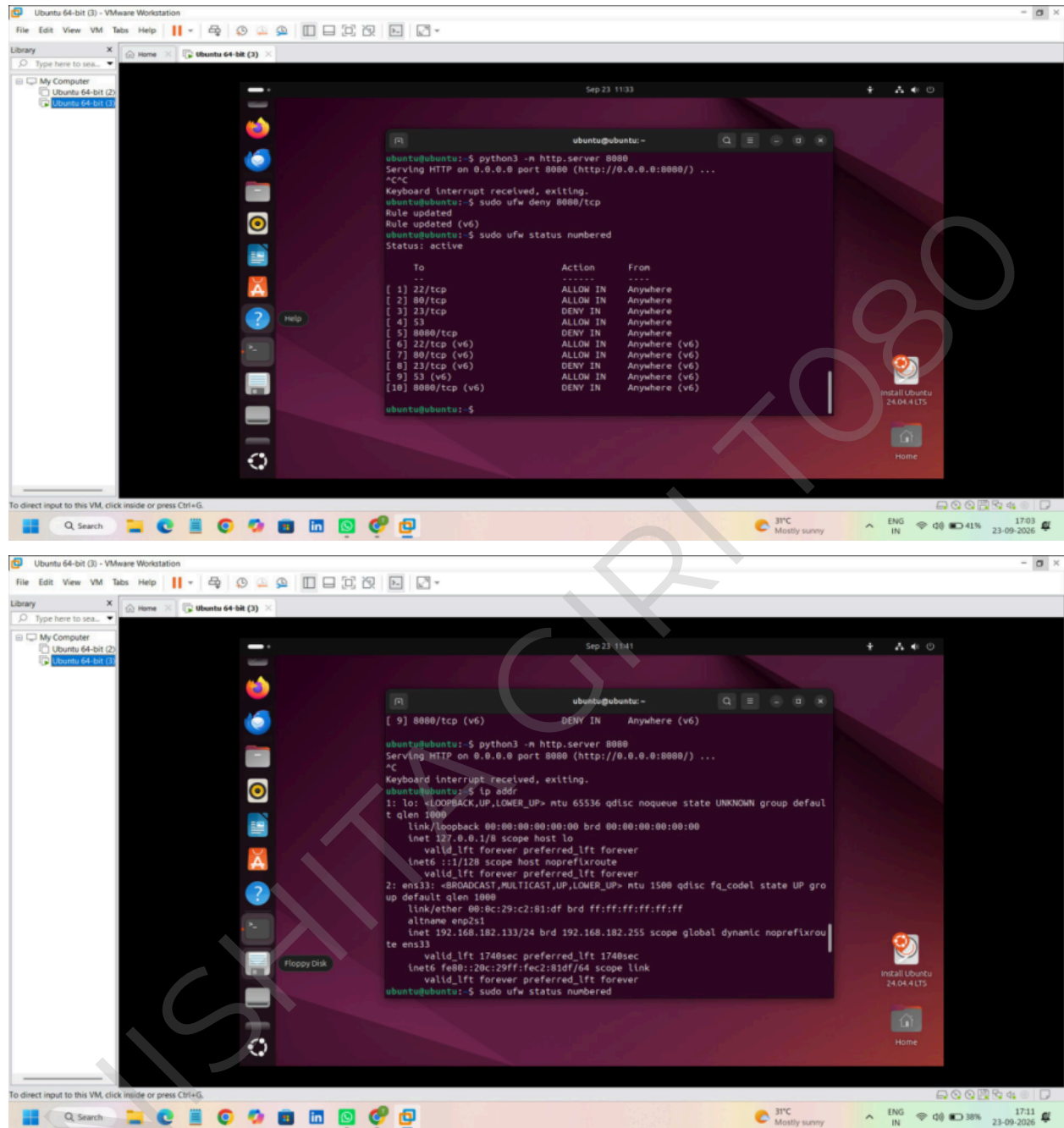
SHETH LUJ AND SIR MV COLLEGE

Step 9: Test the Firewall



SHETH LUJ AND SIR MV COLLEGE

Step 10: Test Blocking



SHETH LUJ AND SIR MV COLLEGE

